TITAN

A multimodal whole-slide foundation model for pathology

Reusable slide representations

Visual self-supervision meets regional and whole-slide language.

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Nature Medicine 31, 3749–3761 (2025)
5 November 2025 · DOI: 10.1038/s41591-025-03982-3

335,645

whole-slide images · 20 organs

423,122

synthetic ROI-caption pairs

182,862

slide-report pairs

Whole slides need context

Gigapixel images, limited labels

A slide contains a long, variable sequence of patch features.

Small cohorts constrain learning

Rare conditions particularly need transferable representations.

One embedding, multiple tasks

Classification, prognosis, and retrieval.

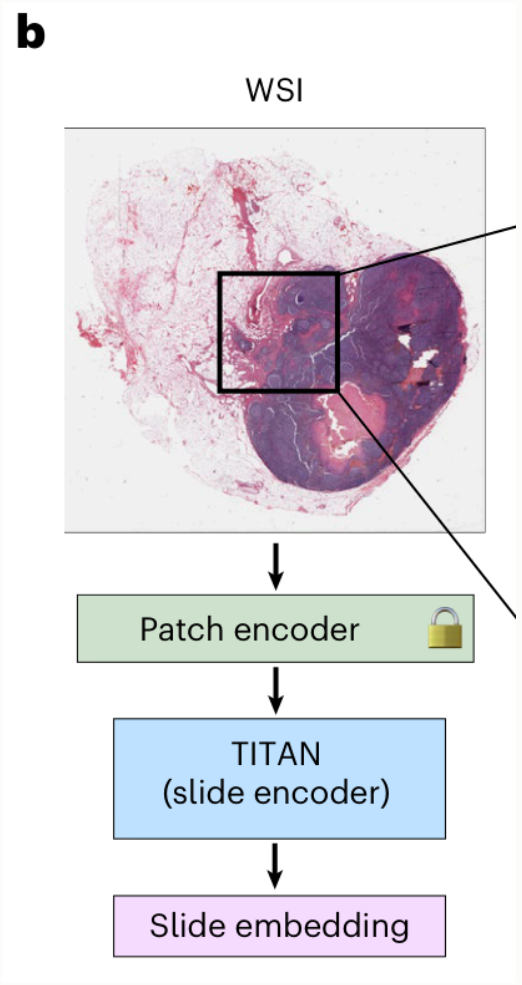


Fig. 1b · A patch encoder feeds a contextual slide encoder.

What should a slide model learn?

01

Tissue context

Can patch-level SSL transfer to whole slides?

→

02

Language alignment

Do regional captions and reports add complementary information?

→

03

Transfer

Can frozen embeddings serve limited-data tasks?

> Evaluate visual quality, language capabilities, and rare-cancer retrieval separately.

Preserve the spatial feature grid

Train on features, retain tissue layout

Pre-extracted CONCHv1.5 features become the slide Transformer’s input.

01

Tissue patches

512 × 512 pixels at 20×

→

02

Patch features

768 dimensions · fixed patch encoder

→

03

Spatial grid

Original patch positions preserved

→

04

Slide embedding

Contextual TITAN representation

> All three TITAN training stages operate in feature space.

Fig. 1b,c; Methods: vision pretraining

Three stages build TITAN

Visual learning comes first

Then detailed morphology and clinical language align the representation.

01

1 · Visual SSL

ROI feature crops → iBOT → $TITAN_V$

→

02

2 · ROI captions

423,122 region-caption pairs → CoCa

→

03

3 · Slide reports

182,862 slide-report pairs → TITAN

> $TITAN_V$ names the vision-only model; TITAN names the full model.

Fig. 1c,d; Results: scaling SSL

Learn from multiple regional views

One region, multiple views

Start with a **16 × 16** feature grid covering 8,192 × 8,192 pixels.

Two global, ten local crops

14 × 14 global grids; 6 × 6 local grids.

Augment in feature space

Horizontal/vertical flips and feature posterization.

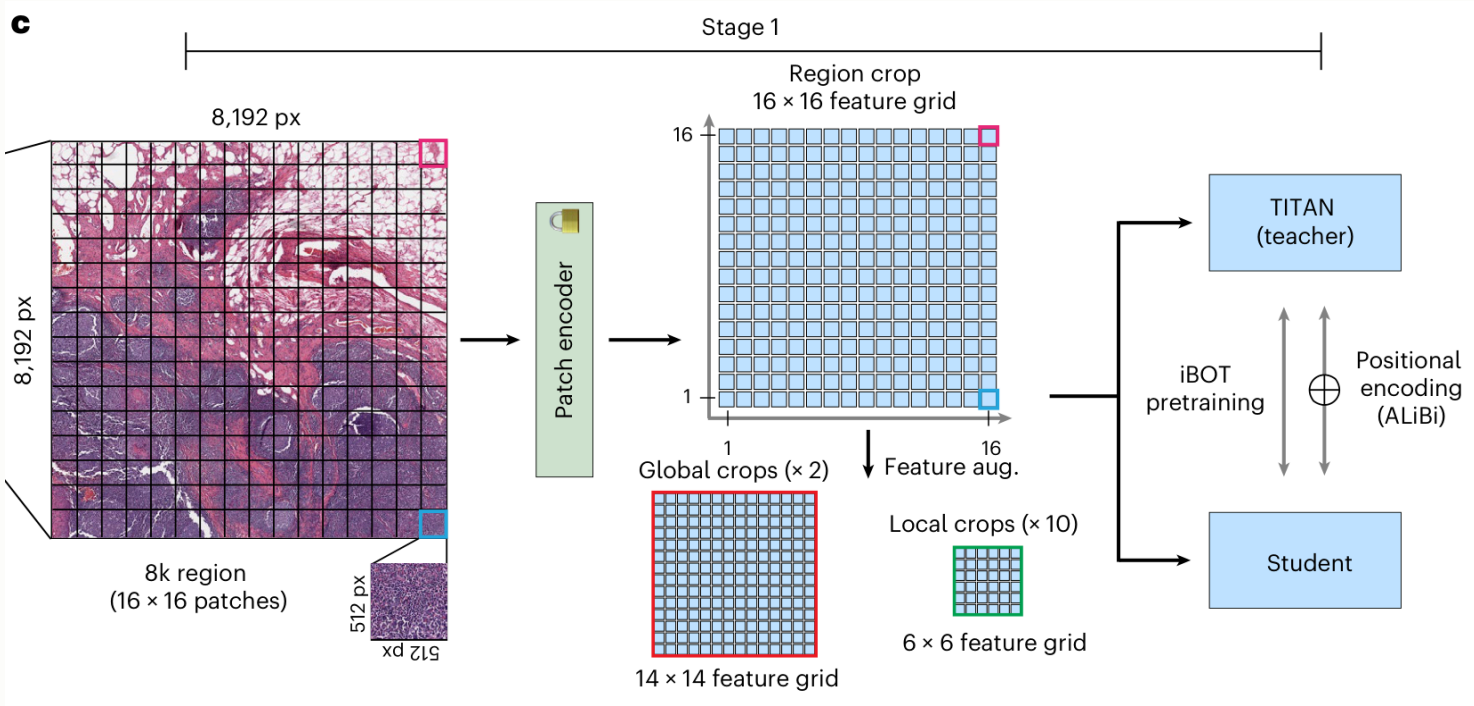


Fig. 1c · iBOT teacher–student learning on spatial features.

Fig. 1c; Methods: vision pretraining

Captions provide morphological detail

Select representative tissue before describing it

An 8,192 × 8,192 ROI supplies diverse visual evidence.

01

64 patches

1,024 × 1,024 pixels each

→

02

16 clusters

K-means; one patch per cluster

→

03

PathChat

Describe the selected patches

→

04

Qwen2 rewriting

Diversify language

> Stage 2 supplies **423,122** ROI-caption pairs; stage 3 adds curated slide-specific reports.

Methods: synthetic caption generation; slide-report curation

Two interfaces support two uses

A shared visual backbone feeds separate poolers

The CoCa interface supports alignment and report generation.

TITAN visual backbone

6 Transformer layers · width 768

→

1 contrastive query

Global visual embedding ↔ text embedding

Zero-shot classification and retrieval

→

128 reconstruction queries

128 visual tokens → multimodal text decoder

Pathology report generation

Text encoder and decoder: 12 layers each, width 768, hidden width 3,072. Initialized from CONCHv1.5; all three components fine-tuned.

Methods: vision-language continual pretraining · diagram redrawn from text

ALiBi adds a spatial distance penalty

Nearby patches receive a stronger attention prior

Train on regions; extrapolate to larger slide contexts.

$$\text{softmax}\left(q_i k_j^\top - m\sqrt{(i_x - j_x)^2 + (i_y - j_y)^2}\right)$$

$q_i\ k_j^T$

Content similarity
query and key vectors in $\mathbb{R}^d$

Euclidean distance

Separation of patch positions
20× coordinates divided by 512

m

Predefined slope
specific to each attention head

> Longer-context inference is supported; full-slide contextual coverage remains a limitation.

Zero-shot prediction compares embeddings

Choose the closest class text

No task-specific classifier is fitted for this evaluation.

$$\hat{y}_i = \arg\max_c u_i^\top v_c$$

u_i

ℓ_2-normalized slide embedding

v_c

ℓ_2-normalized class-text embedding

Ensemble the prompts

Multiple wordings reduce prompt sensitivity.

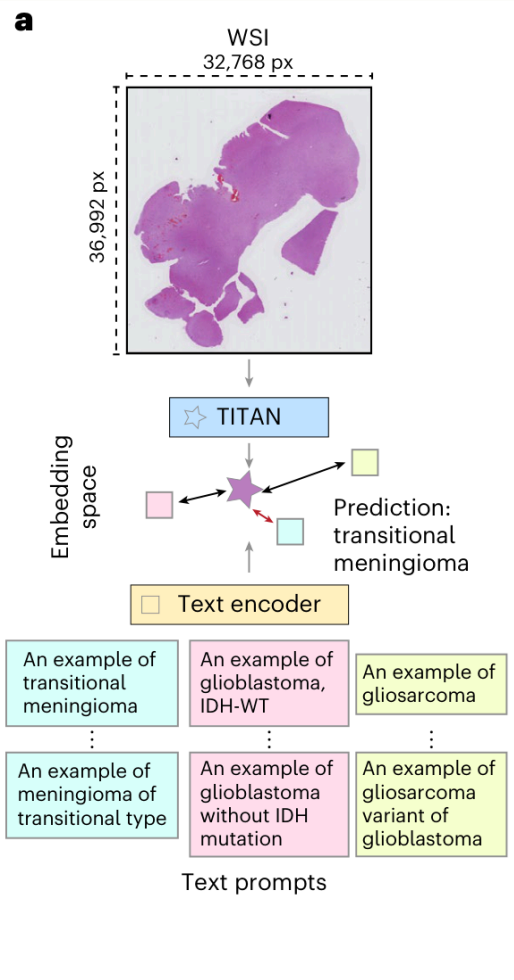


Fig. 3a · Select the class with the greatest cosine similarity.

Evaluate transfer across clinical tasks

Separate pretraining from evaluation

Mass-340K spans **20 organs** and multiple stains and scanners.

Test several uses of frozen embeddings

Morphology, molecular status, survival, and low-data learning.

Quantify uncertainty

Five folds or 1,000 bootstraps; few-shot results over 50 sampled runs.

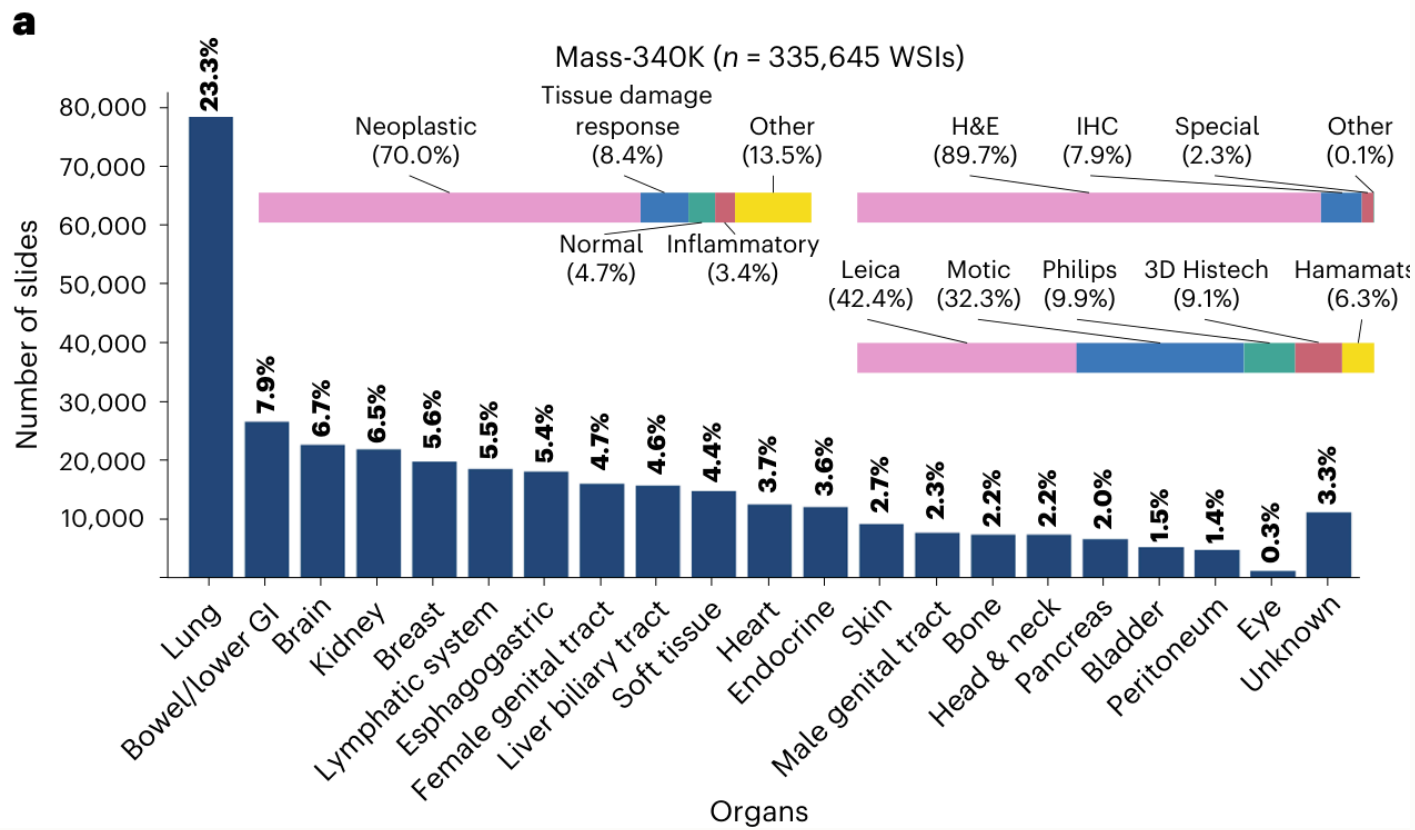


Fig. 1a · Mass-340K contains 335,645 WSIs.

Fig. 1a; Methods: evaluation settings and datasets

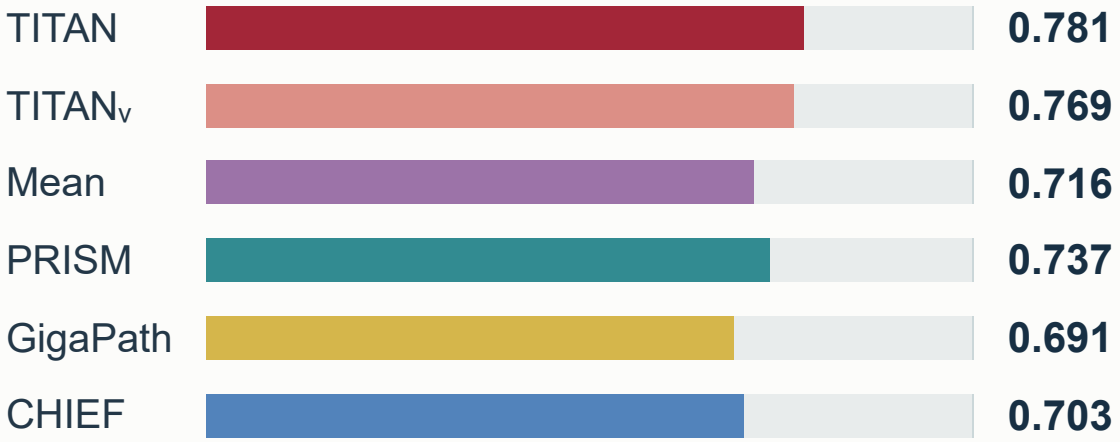
TITAN leads the morphology comparison

Language alignment improves the aggregate score

TITAN exceeds $TITAN_V$ and the evaluated baselines numerically.

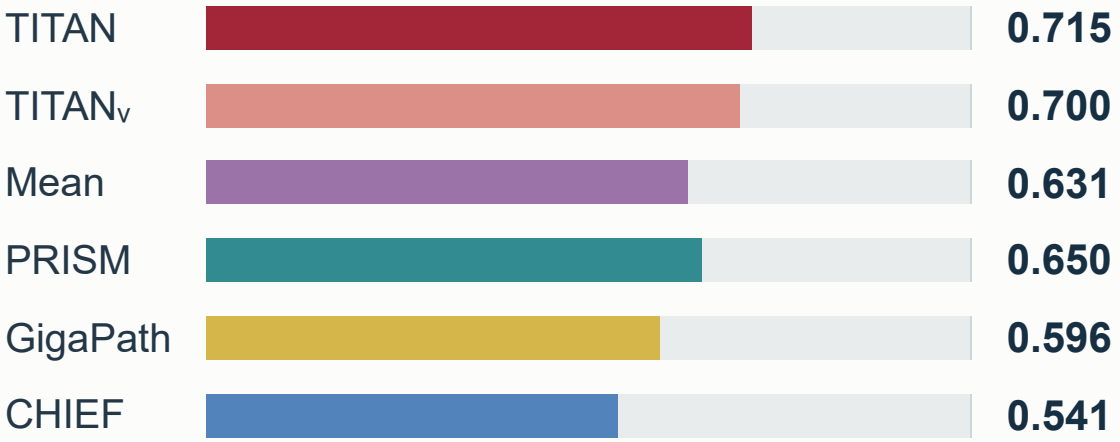
Morphological classification

14 tasks · mixed balanced accuracy / AUROC · scale 0–1



Challenging morphology

4 tasks · balanced accuracy · scale 0–1



The four challenging tasks are a subset of morphology. Different patch encoders and pretraining datasets limit controlled attribution.

Fig. 2c · Values transcribed from printed aggregate labels

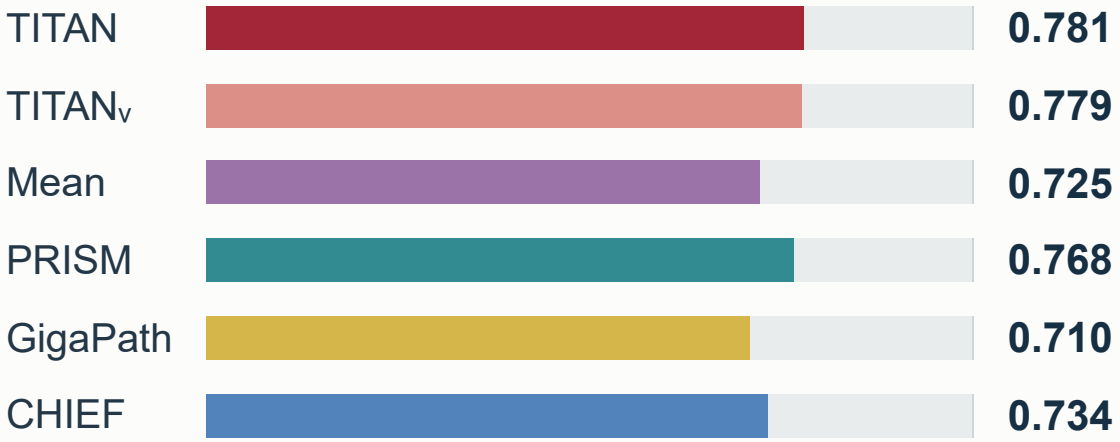
Numerical leads need statistical context

Molecular gains are not uniformly significant

The molecular comparisons with PRISM and $TITAN_V$ do not reach $P < 0.05$.

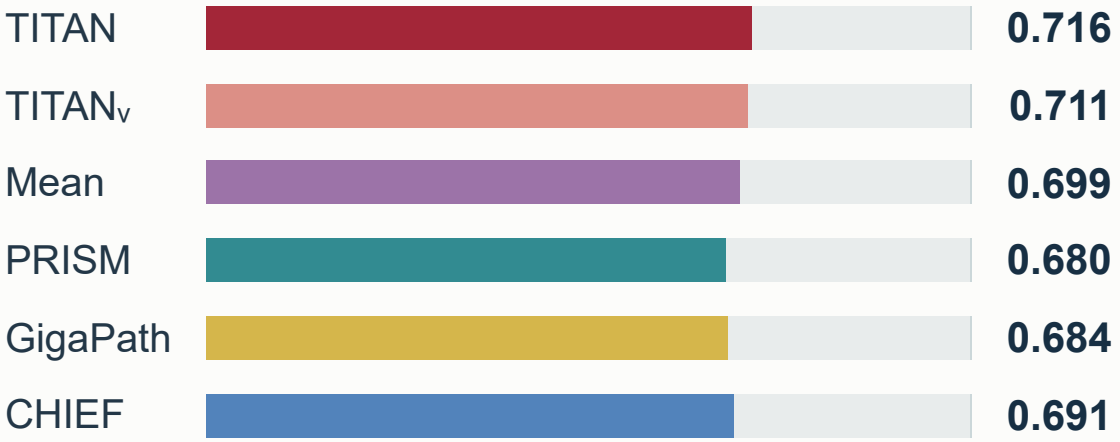
Molecular classification

39 tasks · AUROC · scale 0–1



Survival

6 tasks · c-index · scale 0–1



Molecular: TITAN vs PRISM P = 0.060; vs $TITAN_V$ P = 0.420. Hierarchical beta GLMM, two-sided Wald tests, Tukey correction.

Fig. 2c; Methods: statistical analysis

Language alignment enables retrieval

Retrieve in both directions

Aligned slide and text embeddings share a searchable space.

+10.5%

report → slide Recall@K

+20.5%

slide → report Recall@K

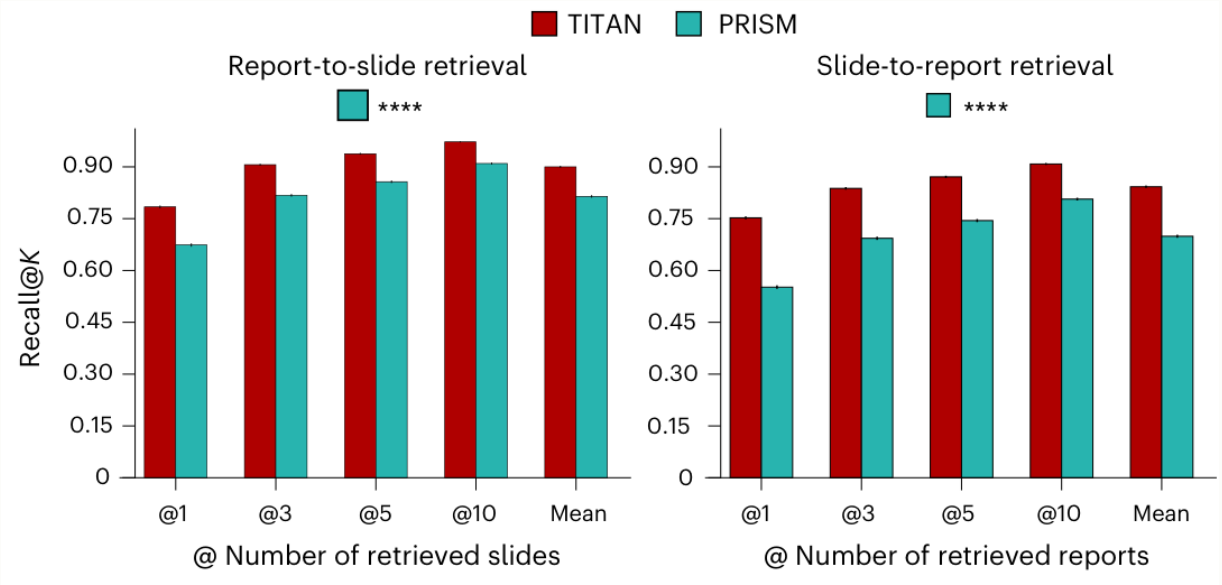


Fig. 4d · 10,108 TCGA slide-report pairs; K = 1, 3, 5, 10.

Authors' relative gains over PRISM, averaged over K; these are not percentage-point differences. Retrieval success is class based, not exact report identity.

Results: cross-modal retrieval; Fig. 4d; Methods

Ablations favor spatial context

Position and depth matter

Six-layer $TITAN_V$ with ALiBi gives the strongest tested configuration.

ALiBi vs absolute position

+1.89% reported performance gain.

Six vs deeper or shallower

+1.16% vs 12 layers; **+1.74%** vs 4 layers.

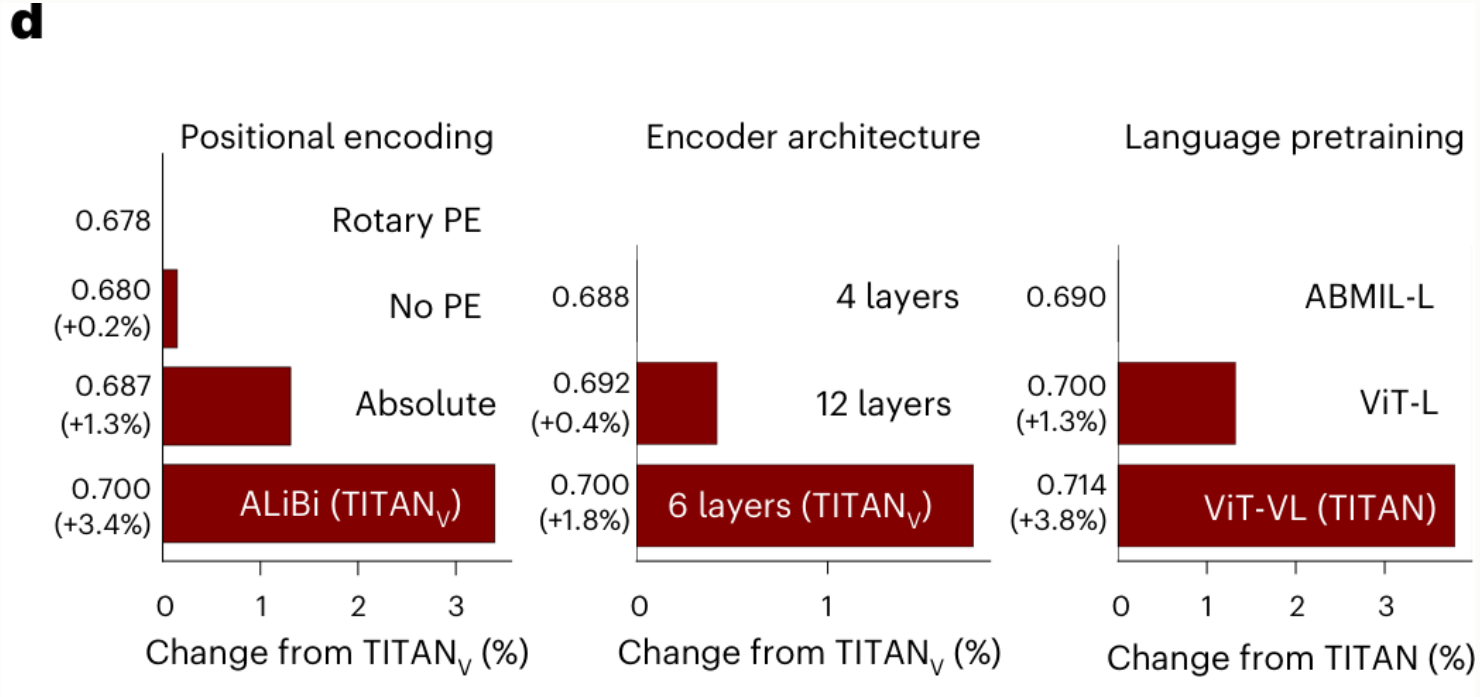


Fig. 2d · Four challenging subtyping tasks.

Relative gains reported in the Results text. Rounded chart labels use a different reference direction; do not reinterpret them as percentage points.

Fig. 2d; Results: design ablations

Reports have the largest zero-shot effect

Remove one pretraining stage at a time

Compare mean balanced accuracy across four challenging subtyping tasks.

Full model: 0.599

Without ROI captions: **0.577**; without reports: **0.561**.

Vision SSL changes this metric less

Without stage 1: **0.598**. This does not erase its visual-transfer contribution.

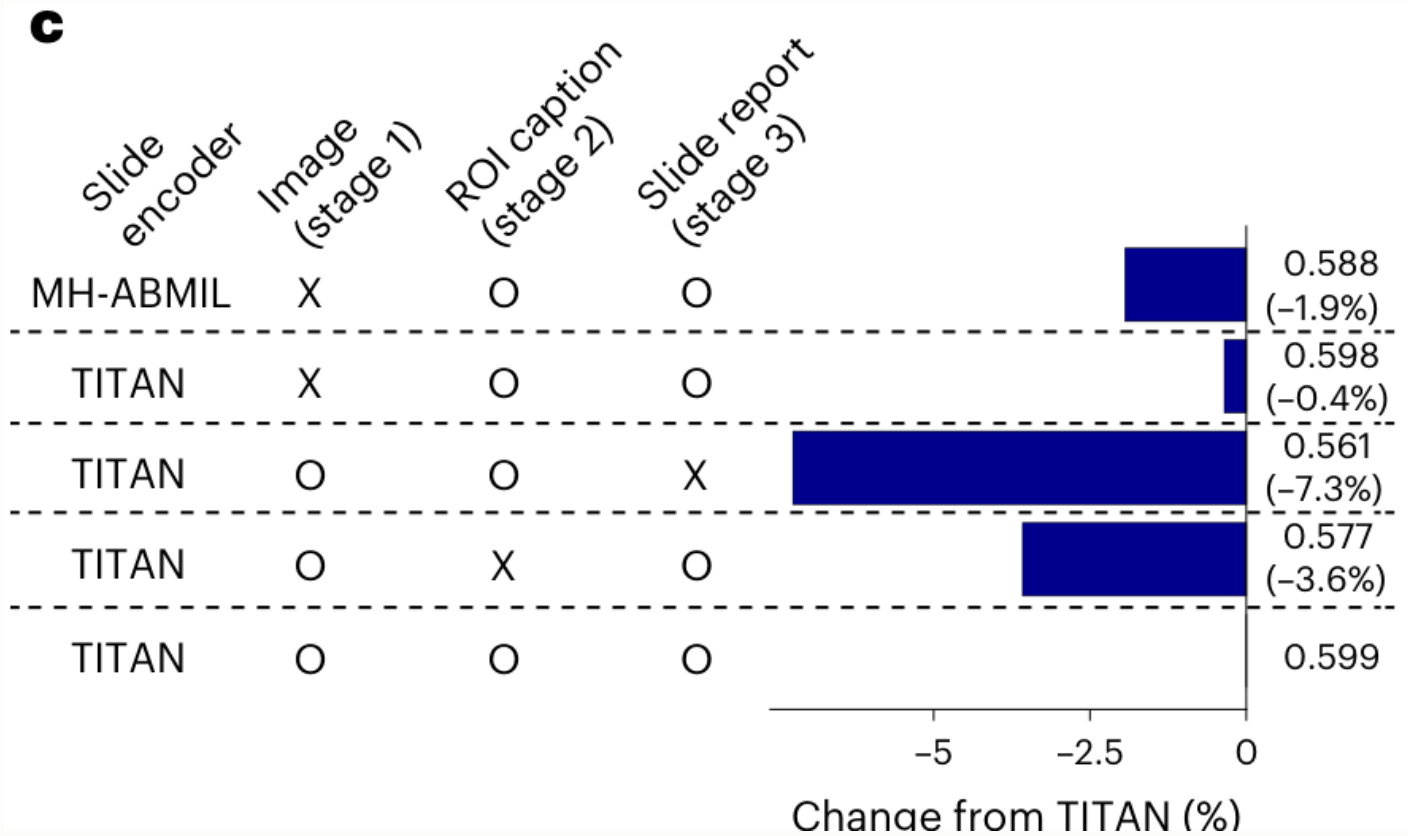


Fig. 3c · Stage removal changes zero-shot performance.

Printed changes: −0.4%, −3.6%, −7.3% against TITAN; retain the figure's rounded values. MH-ABMIL: 0.588 (−1.9%).

Fig. 3c; Results: language-aligned TITAN

One report example shows diagnostic agreement

Cervical squamous carcinoma

TITAN agrees with the reported diagnosis in this selected example.

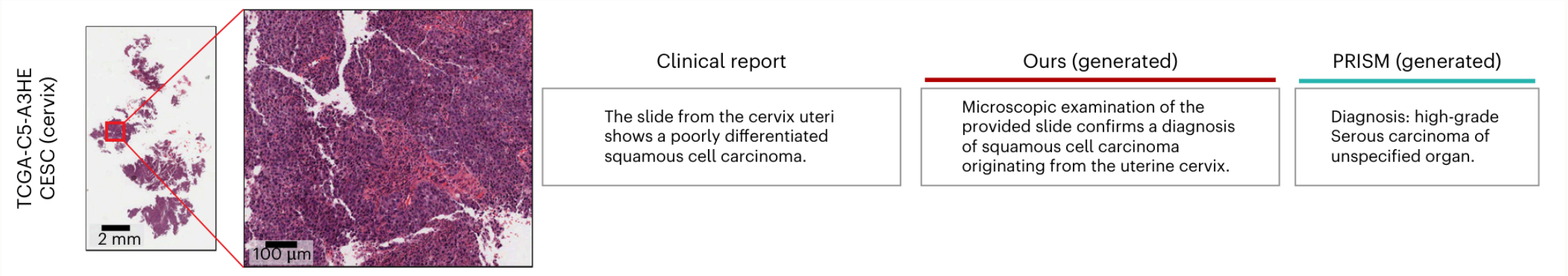


Fig. 3e · TCGA-C5-A3HL: slide, clinical report, TITAN, and PRISM.

Clinical report

Poorly differentiated squamous carcinoma

TITAN

Cervical squamous carcinoma

PRISM

High-grade serous carcinoma; organ unspecified

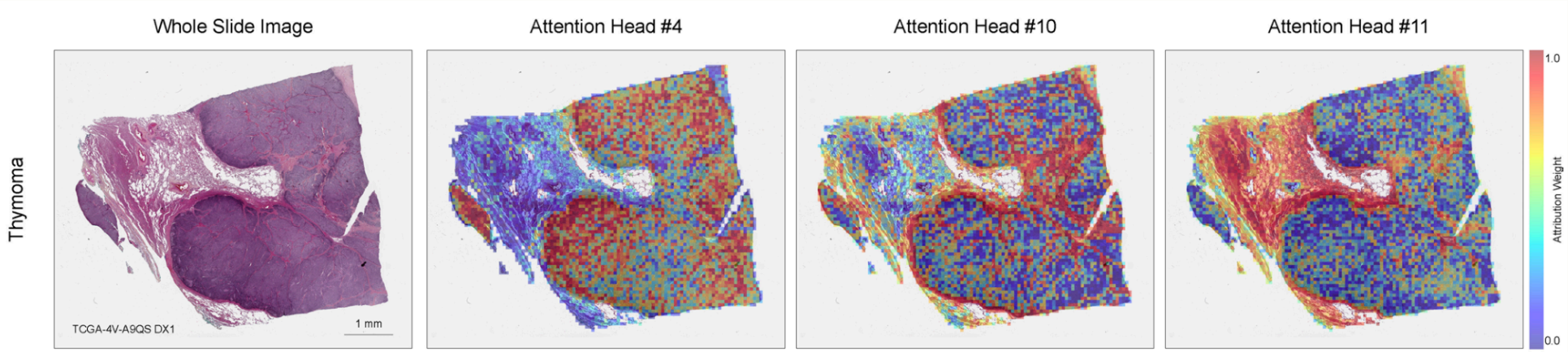
Diagnostic summaries are paraphrased. Author-selected success; not an exhaustive error analysis or clinical validation.

Fig. 3e; report-generation results

Attention heads emphasize different tissue

Different heads attend to distinct morphology

The authors identify tumor, stroma, and nontumor patterns.



Extended Data Fig. 5 · Thymoma: whole slide and heads 4, 10, 11.

Most heads

Dense tumor

Head 10

Tumor-adjacent stroma

Head 11

Nontumor / tissue-specific morphology

Visual self-attention is descriptive evidence; cross-modal attention maps were not reported.

Retrieval follows morphological similarity

A paraganglioma query finds related tissue

The highest displayed similarity is another paraganglioma.

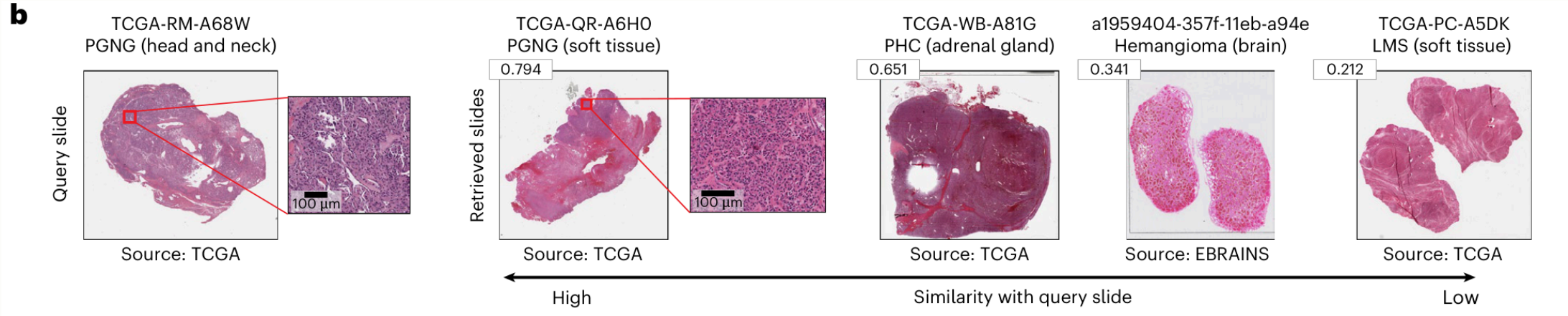


Fig. 4b · Query and four representative retrieved slides.

0.794
Paraganglioma

0.651
Pheochromocytoma

0.341
Hemangioma

0.212
Leiomyosarcoma

Cosine similarities for a selected case. These four examples are not a complete ranked-result or failure analysis.

Fig. 4b; Extended Data Fig. 8

A compact slide encoder is promising

48.5 million slide-encoder parameters

TITAN$_V$: 42.1M; GigaPath: 86.3M; PRISM: 99.0M.

Reduce sequence length

512² instead of 256²-pixel patches gives **4× fewer tokens** for the same area.

Runtime still needs measurement

Parameter counts alone do not establish latency or throughput.

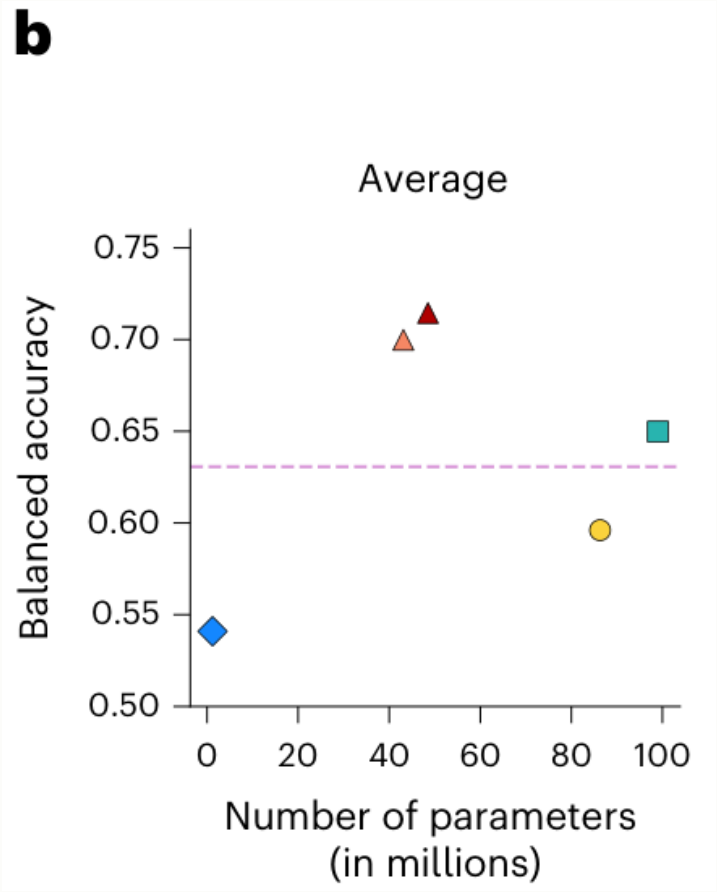


Fig. 2b · Aggregate balanced accuracy vs slide-encoder size.

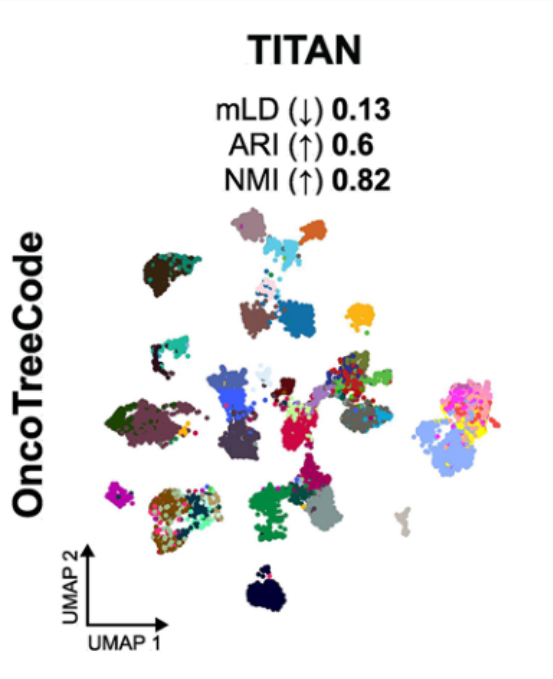
Slide-encoder counts exclude the full patch-plus-language system. Latency, GPU-hours, FLOPs, and comparative peak memory are not reported in the supplied PDF.

Fig. 2b; Results; Methods

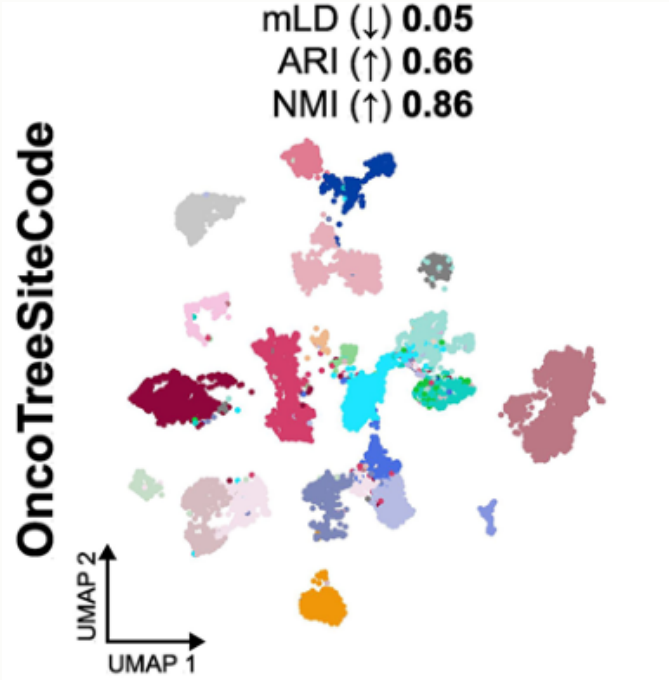
Site mixing does not prove robustness

TITAN embeddings mix submission sites

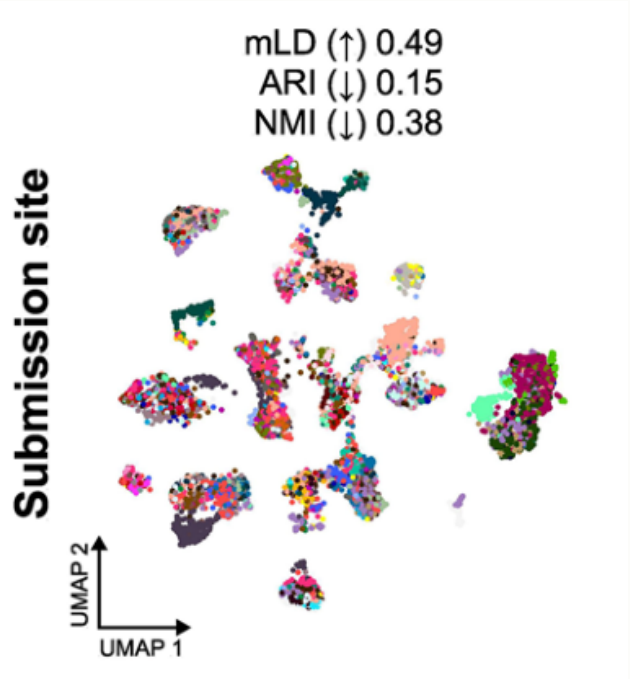
Biological organization remains visible in the same TCGA-OT cohort.



Tumor type · OncoTreeCode



Organ · OncoTreeSiteCode



Submission site

Extended Data Fig. 4: TITAN column, n = 11,186. Site mixing is observational; scanner and processing signals remain an author-stated concern. CHIEF includes TCGA in pretraining.

Extended Data Fig. 4; Discussion pp. 3757–3758

Context and language improve transfer

01

Visual context

Feature-space SSL and spatial attention learn reusable slide embeddings.

→

02

Complementary language

ROI captions add morphology; reports add diagnostic context.

→

03

Broad evaluation

Benefits span probing, zero-shot tasks, and retrieval.

Future directions proposed by the authors

Scale: more slides, captions, and model capacity

Representation: better patch features, positional encodings, and report structure

Translation: robustness and additional pathology domains

Cross-domain applications and routine clinical use are proposed directions, not demonstrated outcomes.

Discussion; Figs. 2–4